Appendix 30.C: Cordless Telephony – DECT

The DECT system is by far the most widespread *cordless* communications system. While cordless phones seem like an ”old-fashioned” application for wireless, they are still a $ 5 billion market, and both households and enterprises use them to a significant degree. Furthermore, it is instructive to explore the difference between the cellular and cordless standards.

While DECT has made some forays into low power consumption and ultra-high data rates (see Sec. 30.C.1.1), its main market is - at least at the time of this writing - still cordless telephony. Due to this reason (and because example standards for the other applications are discussed in Chapters 31-34), this Appendix focuses on the voice telephony applications and the physical layer basic MAC layer functionality that serves it. In other words, the point of this appendix is not to provide a complete description of the DECT standard, but rather to exemplify how a (relatively simple) cordless standard can be constructed, and how it differs from a cellular standard.

30.C.1 Introduction

30.C.1.1 History

Basic cordless telephony, as discussed in Section 1.2, provides a connection between a (portable) UE and a BS; in the case of enterprise cordless systems, there can be multiple BSs connected together into what looks like a ”mini-cellular” system. The very first cordless phones, introduced in the early 1980s, were analog, enabling unprotected communications between handsets and BSs that were at most a few hundred meters away. The BS served as an interface to the public telephone network or to the private branch exchange. Around 1990, different digital cordless phone standards were developed, which offered a larger variety of services and options than the analog versions.

Since for cordless systems communication is generally between one BS and one UE from the same manufacturer, which are usually sold as a set, the need for standardization was less urgent than in the cellular realm. However, the application in PABXs shows clearly the benefits of standardization. The most important of the cordless standards is the DECT standard, which has been developed by the European Telecommunications Standard Institute (ETSI). Its first incarnation was published in 1992, under the name of Digital *European* Cordless Telecommunications; the name was later changed to *Enhanced* to reflect the global usage. DECT also fulfilled the requirements of IMT-2000, and was a 3G “mode” recognized by the ITU.

Further improvements of the standard were published in 2007 under the name of New Generation DECT (NG-DECT). One key modification was the introduction of the DECT Packet Radio Service (DPRS), which as the name indicates, enables packet radio communications over DECT links. One of its core motivations was to make DECT more suited to Voice over Internet Protocol (VoIP) telephony, in particular by supporting the Session Initiation Protocol (SIP) and the H.232 protocol. Furthermore, DPRS also serves for broadband data and audio streaming. Other improvements in NG-DECT are wideband audio with better speech quality, video telephoney capability, and enhanced security enhanced. A new audio codec (LC3plus) with even better quality was added in 2019.

Since 2010, enhancements of the DECT standard have aimed for new applications: the DECT Ultra Low Energy (ULE) standard developed 2012-2014 is intended for home automation and other Internet of Things (IoT) applications. A key advantage over, e.g., Bluetooth and Zigbee, is the use of a less crowded frequency band. The most recent version of the standard is DECT-2020 New Radio (NR), which defines a new air interface that can transmit with up to 1.2 Gbit/s. This was approved by ITU-R as a 5G (IMT-2020) standard. However, at least at the time of this writing, DECT ULE and DECT 2020-NR do not have a significant use for cordless voice transmission, and in the field of data communication are less prevalent than other standards (3GPP NR, WiFi, Bluetooth) discussed in this book. We do not discuss them further in this appendix.

Different regions in the world use slightly different frequency bands (see Sec. 30.C.2.2). In Europe, the 1800 – 1900 MHz band is used; (part of) this band is also used in large parts of Asia and Oceania. North America originally used the 900 MHz and 2.4 GHz ISM bands. Since 2005, it is permissible to operate in the 1920-1930 band; products operating in this band are called DECT 6.0; some NG-DECT products certified with particular operating profiles (so-called CAT-iq 2.0, which enable multiple lines, call transfer, etc.) are called DECT 8.0. These terms are not referring to standards versions (or frequency bands) and are pure marketing terms. Generally, due to different frequency assignments, DECT products produced for one geographic world region must not be operated in another region.

Besides DECT, a number of other cordless standards were developed over the years. In Japan in the 1990s, the Personal Handyphone System (PHS) was not only a cordless system but actually tried to cover the whole country and essentially become a competitor to cellular systems; however, it was replaced by the introduction of 3G cellular systems. In North America, a number of cordless systems using frequency hopping and operating in the 2.45 GHz ISM band were popular for a while, but the interference from WiFi and Bluetooth led to unsatisfactory Quality of Experience (QoE). These systems are now mainly of historical interest.

30.C.1.2 Usage scenarios and basic structure

One of the goals of the development of DECT was to specify a standard that supports a variety of application scenarios. Those include cordless private phones, cordless office phone systems, systems for the public sector (quasi-cellular systems), cordless Local Area Networks (LANs) and wireless access techniques for public networks without mobility support, i.e., Wireless Local Loop (WLL).[1]

A DECT system consists of one or more UEs[2] and one or more BSs. The UEs are connected to the BS via the specified air interface. One BS can serve multiple UEs at the same time. Normally, the BSs are connected directly to the public or private network. However, it is also possible that several BSs may be connected to an optional unit that controls all of them; in that case, this unit establishes the connection to the public network or the private branch exchange. In the case that a system consists of several BSs that are located in different places, a pico-cellular structure of the system may be established. In case that a UE is leaving the coverage area of one BS a *handover* may be performed to another BS associated with the same system. However, in contrast to cellular systems, DECT does not require to assign one particular carrier to each cell, so that no frequency planning is required.

Private cordless phones

The simplest setup for a DECT system in a private environment consists of one UE and one BS. Usually, this BS serves also as the charging station for the UE and is connected to the Public Switched Telephone Network (PSTN) for traditional voice communication, or the internet (for data communication and IP-telephony). Most of the available DECT BSs are capable of supporting several UEs.

In case the BS is connected to the PSTN, calls to the regular landline networks are charged at the normal (landline) rates by the provider. If the BS is connected to the internet, charging underlies the usual internet provider rules; such calls could, e.g., run up against data caps by the provider, or be charged if the internet provider charges per GByte. If multiple UEs are used in the same DECT system, the calls from one UE to the other UE is free of charge, because they do not involve the outside network.

Cordless office communication systems

Cordless office communication systems usually support the same services as regular private branch exchanges. A pico-cellular setup is often used to provide enough capacity and to maintain coverage over bigger office complexes or commercial buildings. The setup normally consists of a regular wired switching unit and an additional control entity, which might be either standalone

[1] Due to various deregulation issues this latter application never became practically important (see also Sec. 1.1).

[2] The DECT standard uses the terms Radio Fixed Part (RFP) and Portable Part (PP) for BS and UE, respectively.

or built into a BS. This control unit supports the mobility management functions of the cordless system, for example, it enables the handover between different BSs. Regular (landline) devices, such as office phones, can also be connected to the private branch exchange.

Cordless phone systems in the public sector - quasi-cellular systems

The DECT standard supports setting up a cluster of BSs that provide access to the PSTN or internet. This cluster consists of pico-cells. As the range of the DECT BSs is rather limited, and the available bandwidth may be smaller, the density of BSs per coverage area must be bigger than for a cellular system. This setup was mostly envisioned for *hotspots*, like airports, train stations or even entire city centers, with users having a contract with a particular provider.

ETSI developed specifications for Cordless Terminal Mobility (CTM), to enable UEs that can operate in a private system, an office setup or in the public sector. Those devices are able to automatically access one of the services, depending on the availability and the user preferences.[3] Furthermore, there was a standard specifying the cooperation between DECT and GSM, the *GSM Interworking Profile* (GIP). The DECT BSs may therefore be connected to the GSM MSCs. All the provisions mentioned above proved to be, however, rather unsuccessful - DECT did not gain much acceptance for quasi-cellular systems.

Cordless LAN

DECT supports the integration of voice and data communications over the same air interface. Much of the work in the standardization since the year 2000 has focused on such data communications, as described above in the historical overview.

Wireless Local Loop

WLLs or Radio Local Loops (RLLs) are radio systems replacing the wired connections on the so-called *last mile*, i.e., the connection between the end user and the local switching center (see also Chapter 1). WLLs are interesting in countries where no sufficient wired infrastructure is available and services should be established fast. In the US and Europe, WLLs were interesting for providers that wanted to enter the deregulated market[4] at the end of the 1990s, however this market essentially vanished due to a combination of regulatory and economic developments, see Sec. 1.1.

30.C.1.2 DECT Application Profiles

To support different access methods, various *profiles* have been specified. The DECT profiles contain additional specifications which define how a DECT device adapts the signaling to the requirements of the accessed networks. These include:

- Generic Access Profile (GAP)
- Corldess Terminal Mobility (CTM) Access Profile (CAP)
- DECT Packet Radio Service (DPRS)
- Open Data Access Profile (ODAP)
- DECT Multimedia Access Profile (DMAP)
- Radio in the local loop Access Profile (RAP)
- Multimedia in the local loop Access Profile (MRAP)

and finally interworking profiles with ISDN, GSM, and UMTS.

[3] For example, the UE tries to use the privately owned BS as much as possible. However, once the coverage of this BS is left the UE accesses the public DECT network of a particular provider.

[4] Until the 1990s, most of the European countries had a single state-owned provider for the regular telephone services. Similarly, the US had a "regulated monopoly" until the mid-1980s.

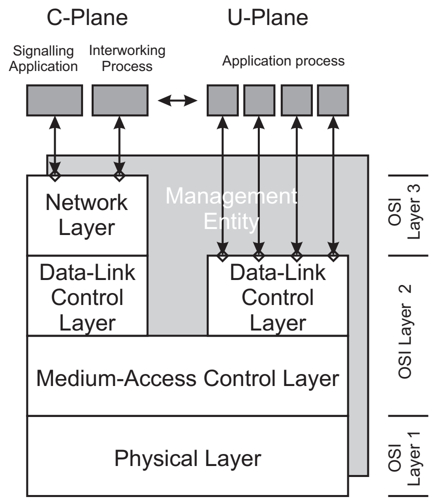

Figure 30.1: DECT protocol stack.

30.C.1.3 The DECT protocol stack

Figure 30.1 illustrates the DECT protocol stack. DECT specifies the lower three layers of the OSI model: the PHysical Layer(PHL), the Data Link Layer (DLL) and the network layer, where the DLL is further divided in the *Medium Access Control* (MAC) layer and the *Data Link Control* (DLC).

The functionality above the MAC layer is divided into a *control plane* and a *user plane* (C-plane and U-plane). The C-plane handles the controlling and signaling functions (addressing, frame control, error control, etc.), whereas the U-plane is concerned with the actual user data transmission; for the C-plane a unicast (point-to-point) and broadcast service is defined. The idea behind the different planes in the DLC is that the transmission of user and controlling data has to be ensured by different means. The network layer specifies (among others) functions for the link control, call control, messaging services, and mobility management. The Lower Layer Management Entity (LLME) contains procedures that concern more than one layer. Finally, the standard specifies (i) identities and addressing schemes, (ii) security features, and (iii) speech and audio coding. The description below will mainly focus on PHY, MAC, and DLC.

30.C.2 The physical layer

30.C.2.1 Overview

The original DECT standard describes a fairly simple system. Table 30.1 gives an overview of the most important parameters, see also [Tuttlebee 2012], [ETSI 2022]. These specifications were considerably extended over time, buth the basic version still forms a subset of the specifications and often forms the core of low-cost handsets.

30.C.2.2 Employed spectrum

The core frequency range for DECT is 1880-1900 MHz, which is a frequency band foreseen exclusively for (unlicensed) voice communications in Europe, and is also the DECT band in much of Asia and Oceania. As indicated in Table 30.1, this allows placement of 10 channels. Modified

Parameter	DECT
Frequency range	1880 ... 1900 MHz
Medium access	FDMA/TDMA/TDD
Number of carriers	10
Channel access procedure	DCS
Distance between carriers	1.728 MHz
Modulation scheme	GFSK (BT = 0.5)
Gross data rate	1152 kbit/s
Slots per time frame	24 full slots
Duration of each time frame	10 ms
Peak transmission power	250 mW
Voice encoding	32 kbit/s ADPCM

Table 30.1: DECT parameter

frequency bands are assigned in other countries: Japan uses 1893-1906 MHz; most of Central and South America uses 1910-1930, and the US and Canada use 1920-1930 MHz, though not on an exclusive basis (other voice systems can operate there as well).

30.C.2.3 Multiple access and duplexing

FDMA

DECT employs a combination of FDMA/TDMA with *Time Division Duplex* (TDD) for the multiple access on the air interface. The used frequency span from 1880 to 1900MHz is divided into ten FDMA bands, centered on:

$$f_c = 1897.344 \text{ MHz} - i \cdot 1728\text{kHz} \qquad \text{with} \quad i = 0, 1, ..., 9 \tag{30.1}$$

so that the spacing between carriers is 1.728 MHz, and guardbands at the upper and lower end of the frequency band reduce the out-of-band interference. The carrier should have no more than 50kHz offset from this definition during transmission.

TDMA

The time axis is segmented into TDMA frames of duration 10ms, during which a total of 11520 symbols are transmitted - hence the bit rate of 1.152 Mbit/s when binary modulation is used; for higher-order modulation the symbol rate stays unchanged, and the bit rates scales up accordingly. Each frame consists of 24 timeslots, each of which thus contains 480 symbols. The first 12 slots are used for the downlink, whereas the second 12 slots serve for the uplink. Every voice communication radio link has a duplex channel (uplink and downlink) assigned. Thus, up to 12 connections can be supported per carrier. Every BS and every UE can access all of the 10 carriers. All in all, 120 duplex links are possible, see Fig. 30.2. Changes of the carrier frequency are possible from one timeslot to the next. Therefore, connections to the same BS can use different carriers for different timeslots. However, in order to access multiple carriers *simultaneously*, the BS needs multiple transceivers. In case that only one transceiver is implemented at the BS only one carrier can be used in each timeslot. Thus, the number of duplex connections is limited to twelve per transceiver BS.

As stated above, a frame usually contains 24 *full slots*. A full slot is used to transmit regular data packets, *Basic Packet P32*, or short packets for signaling purposes, *Short Packet P00*, see below for definition of the various packet types. Furthermore, a full slot may be divided into two half slots (containing 240 symbols each), which may be used to transmit short data packets. Two full slots my be combined to form a double slot (960 symbols) during which a long data packet may be transmitted. It is also possible to define variable-capacity slots.

Figure 30.3 summarizes the division of a frame in slots. In the *basic connection,* regularly used for the voice transmission, the first twelve slots are used for the downlink and the second twelve

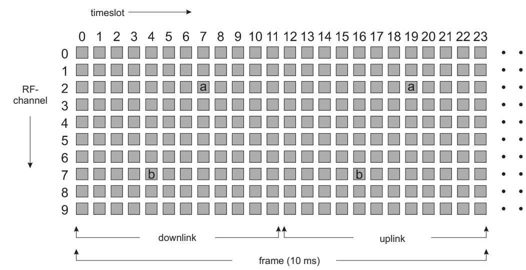

Figure 30.2: Division of the time and frequency in timeslots and subbands.

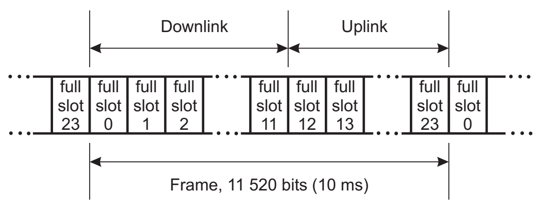

Figure 30.3: Structure of a frame for basic connection. Binary modulation is assumed.

slots are used for the uplink. If the downlink is located on slot number k the uplink uses slot number $k + 12$. *Advanced connections* have a more flexible assignment.

Structure of the different packet types

The user and control data is transmitted in packets. All packets consists of

- *S-field* for synchronization. It consists of 32 symbols; the first 16 of which are alternating $+1$ and -1 and serve as a preamble that can be used for timing and frequency acquisition. The second 16 symbol support the packet synchronization; for their specific values, please refer to the standard [ETSI 2022].
- *D-field* of variable length (depending on the kind of packet, see below), which contains the user and control data. Its composition is explained in Sec. 30.C.3. The D-field may be followed by a Z-field.
- *Z-field* may be transmitted, which repeats the last 4 symbols of the D-field. The receiver compares the two versions of these 4 symbols to recognize interferences.
- *Guard interval*: each packet is shorter than the nominal slot duration (full, half, or double, as applicable), with the remainder serving as guard time, between packets to ensure that packets originating from different UEs are not overlapping at the receiver.

The standard then distinguishes the following types of packets (see also Figure 30.4):

1. *P32*: This packet type is used by regular voice transmission connections, as well as other services. After the 32-symbol S-field, the D-field carries a total of 388 symbols and is composed in the MAC layer. Out of those 388 symbols, up to 320 symbols of the D-field are actually user data. Given the frame duration of 10ms this results in a data rate of 32kbit/s for binary modulation. A Z-field may follow the D field. The duration of a full slot corresponds to 480 symbol durations. A P32 packet is 56 symbol durations shorter than the actual time slot,[5], i.e., the guard time is 56 symbol durations long.

[5]in case that no Z-field is transmitted, it is 60 symbol durations shorter

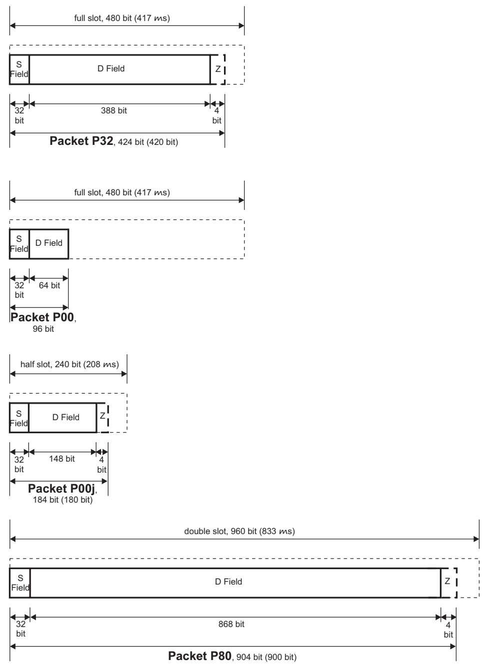

Figure 30.4: The different packet types of the PHY. For the P00j packet, the example of $j = 80$ is shown.

2. *P00*: The P00 packet is used for transmitting brief information for, e.g., control functions. Its D-field has only 64 symbols and it is never send with a Z-field. Therefore, it features a much bigger guard time (384 symbol durations) than the other packets.

3. *P00j*: P00j packets are *variable capacity physical packets* that consist of $100 + j$ (or $104 + j$ if there is a Z-field) symbols.

4. *P80*: P80 packets occupy double slots for transmission. The D-field is 868 symbols long, out of which up to 800 symbols are pure user data. This results in symbol rates of up to 80 kBaud. A Z-field may follow.

Physical channels

A physical channel is defined by a combination of frequency carrier and timeslots. The are denoted as $R(K, L, M, N)$, where K is the index of the full slot in which the transmission of the packet starts, $L = 0$ or 1 indicates whether transmission starts at the full-slot boundary or 240 symbol durations later, M is the index of the frequency channel, and N is an indicator of the BS number (useful for multi-BS systems). Further indices indicate the packet type, whether a Z field is present, and whether the synchronization field is normal-length or extended.

30.C.2.4 Modulation, coding, and transmit power

Modulation format

The basic modulation used in DECT is GFSK with modulation index 0.5, and a time-bandwidth product of the Gaussian filter $B_{\mathrm{G}}T = 0.5$. A logical +1 reflects a positive frequency shift and a logical 0 a negative shift (compare Chapter 10). The specified nominal maximum modulation

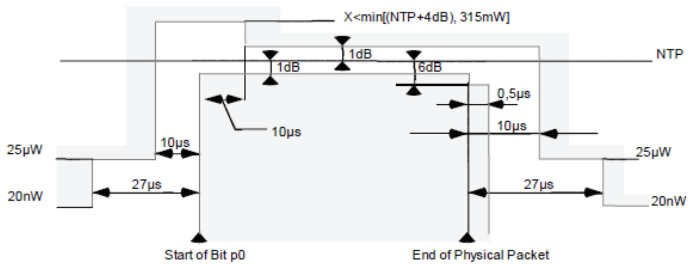

Figure 30.5: Physical packet power-time template. From [ETSI 2022, 30017503].

frequency is 288 kHz, but the implementation may differ within some tolerance range.[6] As the transmission range of the devices is rather limited, the delay spread due to multipath is usually much smaller than the symbol duration. Therefore equalizers are not commonly implemented and no training sequence is foreseen in the standard. The maximum achievable asymmetric data rate is 880 kbit/s. GFSK is defined as modulation scheme 1a, and if it is used it is applied for both control and data field (S and D field). While support for GFSK is not mandatory (there may be devices that only support the below-mentioned $\pi/2-$DBPSK), de facto it is supported by most DECT products.

In later versions of the standard, additional modulation formats with modulation sizes ranging from 2 to 64 was added; the associated data rates range from 880 to 5280 kbit/s. Specifically, in DPRS (see below), modulation schemes 1b, 2,3,5, and 6 are defined, which use $\pi/2-$DBPSK, $\pi/4$-DQPSK. $\pi/8$-D8PSK, 16-QAM, and 64-QAM for the payload data transmission (B+Z field in the D-field), while all of them use $\pi/2-$DBPSK for the control data (the S-field and the A-field in the D-field); these different fields will be explained below. Further modulation combinations are also defined in the PHY standard, but are not required to be supported for DPRS.

Finally, the DECT-NR standard uses a completely different modulation format, based on OFDM. As mentioned in the introduction, DECT-NR is not treated here any further.

Transmission power and receiver sensitivity

The peak transmission power of a DECT device is 250 mW; though this number might be modified according to different regulatory requirements in different regions of the world. As the UE is transmitting only in one out of 24 slots in normal speech mode per frame its average transmission power is about 10 mW. No power control is specified in the standard. The template for the ramping up and ramping down the power before/after a physical packet is shown in Fig. 30.5.

The specifications require a receiver sensitivity of at least -83 dBm to achieve a BER of 10^{-3}. As the used voice encoding (ADPCM) is quite robust, no forward error correction is required and a BER of 10^{-3} results in a sufficient voice quality.

30.C.2.5 Voice encoding

DECT supports four audio modes with different quality: narrowband audio (covering speech signals up to 3.1 kHz), wideband audio (7 kHz), superwideband audio (14 kHz) and fullband audio (20 kHz).

More specifically, DECT originally used *Adaptive Differential Pulse Code Modulation* (ADPCM) with a data rate of 32 kbit/s for voice encoding (see Chapter 24). This voice encoding provides reasonable quality, comparable to the one achieved in ISDN and is rather robust against single bit errors. However, since only frequencies of the voice up to 3.1 kHz are used, the voice quality is still suboptimum.

[6] for the nominal modulation index of 0.5, GFSK is GMSK.

For this reason, later versions of the standard introduced new versions of codecs. NG-DECT uses the G-722 audio codec defined by ITU-T. This is a codec developed in the 1980s, and thus somewhat dated, but it does encode a wider part of the audio spectrum (up to 7 kHz), is low-complexity, and royality-free; it also needs twice the data rate (64 kbit/s); added to this was were the MPEG-4 ER AAC-LD audio codecs with 32 and 64 kbit/s. The most recent versions of DECT use the L3C/L3Cplus audio codec, which was developed by Ericsson and Fraunhofer Institute and is used not only in DECT, but also Bluetooth for both speech and audio transmission.

30.C.2.6 Error correction and detection

Higher-layer information from the user-plane can be transmitted on two different types of channels, called I_N and I_P. The I_N channels are intended for speech and other minimum-delay services, and contain only an error detection using a CRC. The I_P channels can have error detection, retransmission, or error correction with turbo codes (four different B-field formats exist for I_P channels). Higher-layer user-plane control information uses error correction based on ARQ. Higher-layer information from the control plane is sent over the slow duplex channel C_S, or the fast duplex channel C_I. Either of those channels is protected by error correction based on ARQ.

30.C.2.7 Comparison DECT - GSM

Table 30.2 compares several technical parameters of DECT with the ones of GSM. This is of interest since GSM was the cellular standard designed around the same time, and also intended for voice services.

30.C.3 MAC layer and connection setup

30.C.3.1 Tasks for MAC Layer

The MAC layer is responsible for the assignment and access to the physical medium. This includes

- transmission and evaluation of the beacon and broadcast channel at the BS and the UE, respectively.
- hosting the functions for *dynamic channel selection* (see below), for example, the transmission and evaluation of the scanning sequence of the BS.
- matching and multiplexing the logical channels, which may either contain user or control data from higher layers, onto the different fields of the transmitted channel. It may support the association of multiple physical channels with one connection to achieve higher data rates.
- establishment of reliable transmission by Automatic Repeat reQuest (ARQ, see Sec. 13.12) of the control data and - if applicable - the user data.

30.C.3.2 Types of functions

The MAC consists of Cluster Control Functions (CCF) that control more than one cell, as well as Cell Site Functions (CSF) that are dealing with one cell only.

The CCF provides the Broadcast Message Control (BMC), where the broadcast messages carry internal logical channels in the downlink; the Connectionless Message Control (CMC), which distributes information of the Connectionless Bearer Control (CBC), and the Multi-Bearer Control (MBC), which controls the multiplexing of the data associated with a MAC connection between one BS and one UE.

Parameter	**DECT**	**GSM**
Frequency range (Europe)	1880-1900 MHz	880-915 MHz (uplink) 925-960 MHz (downlink)
		1710 – 1785 (uplink) 1805 – 1880 (downlink)
Medium access	FDMA/TDMA/TDD	FDMA/TDMA/FDD
Selection of physical channel	DCS	fixed channel allocation/
		intracell handover/
		frequency hopping
Carrier distance	1.728 MHz	0.2 MHz
Modulation approach	GMSK ($B_{\mathrm{G}}T = 0.5$)	GMSK($B_{\mathrm{G}}T = 0.3$)
Effective frequency usage per duplex speech connection	144 kHz/channel	50 kHz/channel
Cross bit rate on the air interface	1152 kbit/s	271 kbit/s
Symbol duration	0.87μs	3.7μs
Channels per carrier	24 full slots (32 kbit/s user data)	8 full slots (13 kbit/s user data)
Frame duration	10ms	4.6ms
Maximal RF transmission power at the UE	250 mW	2 W
Voice encoding	32 kbit/s ADPCM	13 kbit/s RPE-LTP
Diversity	Antenna diversity (opt.)	Channel coding with interleaving Channel equalization Antenna diversity (opt.) Frequency hopping (opt.)
Maximal cell range	ca. 300 m (outdoor)	35 km
Dedicated physical channels for signaling	not necessary	necessary
Power control	none	mandatory

Table 30.2: Comparison of the technical parameters of DECT/GSM

In the CSF category, there is the Traffic Bearer Control (TBC) that controls one duplex bearer (i.e., two physical channels), the Dummy Bearer Control (DBC) that controls one simplex bearer, i.e., one physical channel, and the CBC, which can control one or two bearers. The Idle Receiver Control (IRC) controls the TXs that are not involved with a current bearer.

30.C.3.3 Types of connections

There are symmetric and asymmetric connections, with the former having the same bandwidth for uplink and downlink, while the latter have service in both directions, but with different bandwidths. There are also multi-bearer connections. For speech communications, symmetric connections are the most important type. Within the category of symmetric connections, there are

- type 1: minimum delay, limited error protection, fixed throughput
- type 2: normal delay, limited error protection, fixed throughput

- type 3: minimum delay, fixed throughput, error detection capability
- type 4: variable throughput, error correction based on ARQ
- type 5: variable throughput, error correction coding

Data rates, delays, etc. are specified in [ETSI 2022 - 300175-3]. Similarly, there is a large number of asymmetric channel types, with varying data rates related to both the number of bearers assigned in the different directions, and the modulation schemes.

30.C.3.4 Multiplexing of the logical channels and backward error correction

Multiframe: 16 of the regular time frames are forming a multiframe. The multiframe structure allows to distribute the data from logical channels (e.g., broadcast) over several transmission packets.

D-field: All transmission packets contain a D-field, whose length depends on the type of packet that is transmitted. The composition and evaluation of this D-field is accomplished in the MAC layer. In the following, we describe the composition of the D-field in the P32 packets, which is illustrated in Figure 30.6. The D-field has a length of 388 bits and is subdivided in the A-field for control data, which contains 64 bit, and the B-field, which consists of 324 bits, mainly for user data. Note that in systems with multiple modulation formats, the standard might enable/prescribe different modulation formats for A- and B-field.

- **A-field:** The A-field consists of 48 bits of signalling information, and is protected by a 16 bit CRC field. It is thus used to provide a permanent signaling channel. This channel is used to transmit control data, like MAC- layer information, higher-layer information, the broadcast channel for paging, BS ID etc, or the dynamic link control.
- **B-field:** The B-field might be used to transmit user data in an unprotected mode or in a protected mode. The *unprotected mode* is used for voice communications. In this case the 324 bits of the B-field contain 320 bit of actual user data and a four-bit X-field, which is a CRC field for the other 320 bits. Note that this X-field is the field which is copied into the Z-field.

  In the *encoded protected* format, the information is interleaved and protected with error correction coding. In the multi-subfield *protected mode* the 320 bits of the B-field are divided into blocks of 80 bits each. Of these 80 bits, 64 are actual user data and 16 bits are a CRC-field (*RA-field*). The X-field is the same as in the unprotected mode. The effective data rate in the protected mode is 25.6 kbit/s instead of 32 kbit/s. In special cases the B-field may also be used for the transmission of signaling data.

30.C.3.5 Packet data and support for transparent IP

NG-DECT specifies the DECT Packet Radio Service (DPRS), which include mechanisms for context control, mobility management, and security. It also forms the basis for the transparent transmission of IP data.

DPRS uses the DECT physical channels, and enables the use of a connectionless protocol on the connection-oriented service provide by the DECT PHY. The links between BS and UEs are asymmetric; the TDD characteristics allow an easy reversal of the transmission characteristics for the different slots; in other words, there is no pre-defined transmission direction but rather it is constantly adjusted to fit the demands of the situation. When GFSK is used, symmetric data speed can be up to $460 + 460$ kbit/s, while asymmetric data speeds of up to $845 + 57.6$ kbit/s can be achieved. When 64-QAM modulation is used, up to 5068 kbit/s can be achieved.

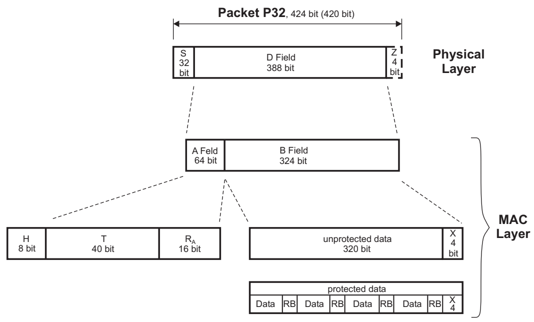

Figure 30.6: Composition of a P32 packet.

The DPRS standard foresees three different categories (cat1, cat2, and cat3) of systems that differ in the data rate, type of packets that are used, and types of services that are enabled. Specifically, cat1 supports symmetric data rates up to 51.2 kbit/s, cat 2 has symmetric data rates of 204.8 kbit/s mandatory (and 358.4 +44.8 kbit/s in an asymmetric connection). Cat 3 provides up to the above-mentioned 845 + 57.6 kbit/s.

For many types of data, the BS will forward the IP packets without modifying them. However, for VoIP, it is usually more efficient, and provides better quality, if the BS extracts the speech information from the IP packets, and forwards the content as "DECT speech".

30.C.3.6 Dynamic Channel Selection (DCS)

DECT uses a dynamic decentralized procedure to select the physical channel for each connection. This procedure is called *Dynamic Channel Selection* (DCS). This procedure is decentralized in the sense that every UE selects individually the carrier for its link. It is dynamic in the sense that the UE or the BS may change the carrier from one frame to the next frame. DCS is a very efficient way of accessing the channels. Each DECT device monitors the channels it does not use to transmit or receive. The UE then identifies the link to a BS that gives the highest signal power, as well as little interference.

30.C.3.7 Broadcast service

Each DECT BS transmits continuously on at least one channel. This is called the *beacon channel*. This beacon contains information about the BS and allows a UE to log on to it. The beacon may be transmitted within a channel of a regular duplex connection or, in the case that there is no regular connection, the BS transmits a *dummy bearer*, which only contains the beaconing signals. The beaconing channel transmits broadcast information. This information is multiplexed over several frames (mulitframe) and includes the BS identity, the supported services, the status of the BS and paging signals for the associated mobiles. The beacon allows a UE to synchronize to the frame and slot timing of the BS. Afterwards, it evaluates the signaling information transmitted via the broadcast channel to check whether the UE has access rights to the BS, if its requested services are supported and whether a duplex channel is available, in case a connection has to be established.

Besides this continuous broadcast, the standard also defines a non-continuous broadcast that allows the UE to acquire more channel information, and a paging broadcast, where the BS requests the UE to set up a connection to the BS.

30.C.3.8 Establishing of a connection

In order to establish a connection from or to the UE, it has to be within range of a BS to which it has access rights and synchronized to the frame timing of this BS. It is always the UE that initiates the channel request, both when the UE initiates a call, and when the BS pages the UE because of an incoming call.

Connection requested by the UE:

The UE selects the best available channel and starts to transmit on it.[7] The BS scans the unused channels sequentially to detect when a UE is accessing one of those. The scan sequence is transmitted via the broadcast channel. Knowing this sequence the UE selects the frame during which the BS observes the chosen channel, to send the first burst. This ensures that the BS immediately detects the channel access of the UE.

Connection requested by the BS:

In case there is an incoming connection request from the network the BS pages the designated UE via the broadcast channel. Upon receiving this paging signal the UE initiates a connection in the same manner as described above.

30.C.3.9 Handover

DECT offers the functionality to hand over a connection to another BS, a so called *intercell handover*, or to switch the channel within a cell, a so called *intracell handover*. The UE is constantly observing the quality of the free channels. In case the channel in use is subject to interferences, for example from neighboring cells, the UE initiates an intracell handover to one of the free channels with low interference. Since the uplink may be corrupted by interferences of which the UE is not aware, also the BS can request a intracell handover via the signaling protocol.

When the UE is in the process of leaving the coverage area of the BS the signal strength and the quality of the link is decreasing. By observing the unused channels the UE detects the beacons of the neighboring BS. When the signal strength of the neighboring BS is bigger than the one of the current BS the UE starts to access a new free channel in this cell and thus initiates a handover. DECT performs a *seamless handover* by first establishing a second channel, only dissolving the original channel after the second channel is operative.

References

ETSI 2022 European Telecommunications Standards Institute, "Digital Enhanced Cordless Telecommunications (DECT)" standards specifications (2022). In particular

- **EN 300 175-1** Common Interface (CI); Part 1: Overview
- **EN 300 175-2** Common Interface (CI); Part 2: Physical Layer (PHL)
- **EN 300 175-3** Common Interface (CI); Part 3: Medium Access Control (MAC) layer
- **EN 300 175-4** Common Interface (CI); Part 4: Data Link Control (DLC) layer
- **300 175-5** Common Interface (CI); Part 5: Network (NWK) layer
- **EN 300 175-6** Common Interface (CI); Part 6: Identities and addressing
- **TR 102 570** New Generation DECT; Overview and Requirements (2007)
- **EN 301 406-1** Harmonised Standard for access to radio spectrum; Part 1: DECT, DECT Evolution and DECT ULE
- **TS 102 527-1** New Generation DECT; Part 1: Wideband speech
- **TS 102 527-2** New Generation DECT; Part 2: Support of transparent IP packet data

Tuttlebee 2012 W. H. Tuttlebee (ed.), Cordless telecommunications in Europe: the evolution of personal communications. Springer Science & Business Media (2012).

[7] Slots during which another connection to the BS is already in use are so called *blind slots*. During these slots the channels on other than the used carriers cannot be used, because the base station cannot support them. Those channels are excluded from the set of free channels out of which the mobile chooses the best one.

30.C.4 Acronyms

ADPCM	Adaptive Differential Pulse Code Modulation
BMC	Broadcast Message Control
CAI	Common Air Interface
CAP	CTM Access Profile
CBC	Connectionless Bearer Control
CCF	Cluster Control Function
CMC	Connectionless Message Control
CSF	Cell Site Function
CT2	Cordless Telephone 2nd Generation
CTM	Cordless Terminal Mobility
DAM	DECT Authentication Module
DBC	Dummy Bearer Control
DPRS	DECT Packet Radio Service
DCS	Dynamic Channel Selection
DECT	Digital Enhanced Cordless Telecommunications
DLC	Data Link Control
DLL	Data Link Layer
DMAP	DECT Multimedia Access Profile
ETS	European Telecommunication Standard
ETSI	European Telecommunication Standards Institute
GAP	Generic Access Profile
GFSK	Gaussian Frequency Shift Keying
GIP	DECT/GSM Interworking Profile
GMSK	Gaussian Minimum Shift Keying
GSM	Global System for Mobile communications
IAP	ISDN Application Profile
IIP	ISDN Interworking Profile
IN	Intelligent Network
IRC	Idle Receiver Control
LLME	Lower Layer Management Entity
LPC	Linear Prediction Coding
MAC	Medium Access Control
MSC	Mobile Switching Center
MBC	Multi-Bearer Control
MRAP	Multimedia RAP
NG-DECT	New Generation DECT
NR	New Radio
ODAP	Open Data Access Profile
OSI	Open System Interconnection
PACS	Personal Access Communications System
PACS-UA	Personal Access Communications System – Unlicensed A
PACS-UB	Personal Access Communications System – Unlicensed B
PCI	Personal Communications Interface
PCS	Personal Communication Systems
PDC	Personal Digital Cellular
PHL	PHysical Layer
PHS	Personal Handyphone System
PWT	Personal Wireless Telecommunications
PWT-E	Personal Wireless Telecommunications – Enhanced
RAP	RLL Access Profile
RLL	Radio in the Local Loop
TBC	Traffic Bearer Control
ULE	Ultra Low Energy
VoIP	Voice over Internet Protocol
WLL	Wireless Local Loop